

To combat these situations where managers incur a loss due to unexpected high trading costs or abandon a strategy that seems unprofitable due again to inaccurate transaction costs, managers are back-testing investment ideas using historical cost index series.

The important part of the cost index is to provide the historical trading cost based on today's market structure conditions not on the historical structure. For example, in the early 1990s the market was trading in 1/8ths or really 1/4s according to the odd-eighths paper of Christie and Stoll (1994). Let's take a look at what has happened since then. The market changed from the 1/8th quoting system to one of 1/16ths (teensies), the SEC order handling rules came into play, decimalization, algorithms, Reg-NMS, growth of electronic trading venues and the proliferation of dark pools, and now having ten, twelve, or more displayed venues compared to only two mutually exclusive exchanges, NYSE and NASDAQ.

All of these regulatory changes have dramatically improved the efficiency of our financial markets and reduced trading costs. Therefore, when we back-test using trading costs our goal is to construct a back-testing cost index series based on the costs that would have occurred historically based on today's market structure and trading environment, not what actually occurred back then. We use today's market structure because we do not want to miss out on an opportunity that would not have been profitable during yesterday's market structure, but given today's market environment the strategy is greatly profitable.

To assist managers resolve these issues and potentially uncover additional investment opportunities, we constructed a cost index based on today's market structure and the historical market conditions. This is shown in [Figure 12.5](#) for US large cap stocks. Our cost index covers the periods from 1991–2012 (22 years). The cost index shows costs gradually decreasing over the 1990s but spiking in the late 1990s (Latam Crisis) and beginning of the tech boom with increased volatility. Costs remained at higher levels with large fluctuations until the tech bubble crash in March 2003. Cost then remained low through the quant crisis and dramatically spiked during the financial crisis, flash crash, and again during the US debt crisis. Quantitative portfolio managers could suffer large losses if they use a constant trading cost such as 20 bp to enter and exit positions. Many times a strategy appears profitable due to lower modeled costs, but these are the strategies that suffer the largest losses when implemented. This cost index could also result in portfolio managers finding that a strategy that would not have been profitable due to the